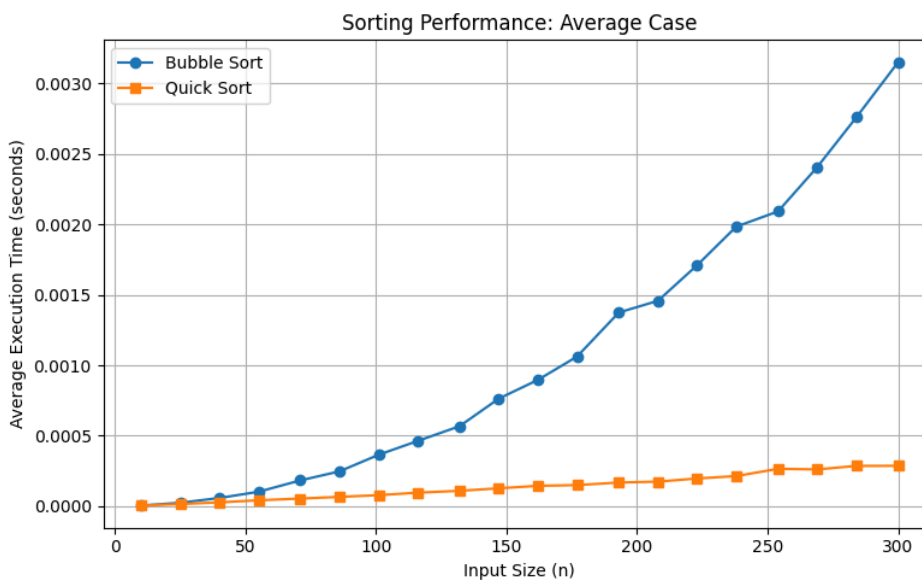
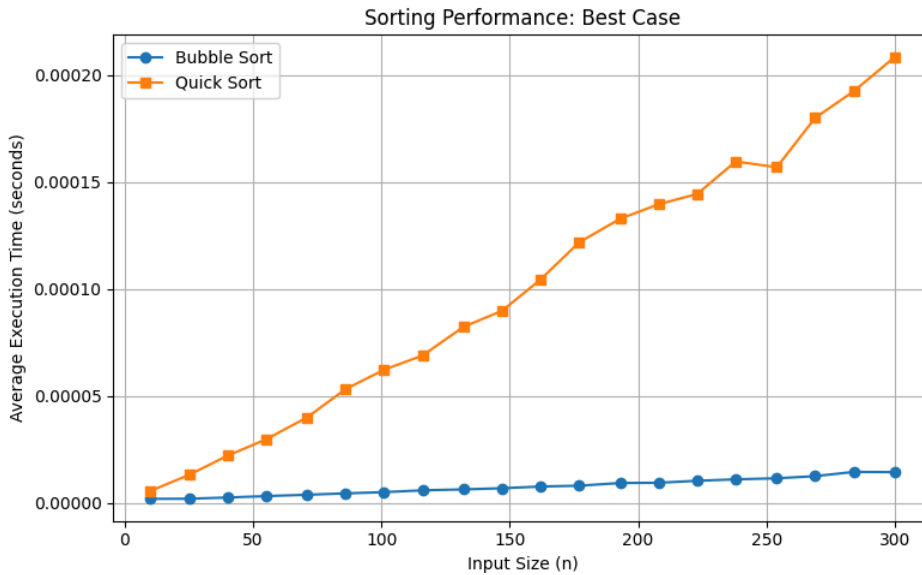
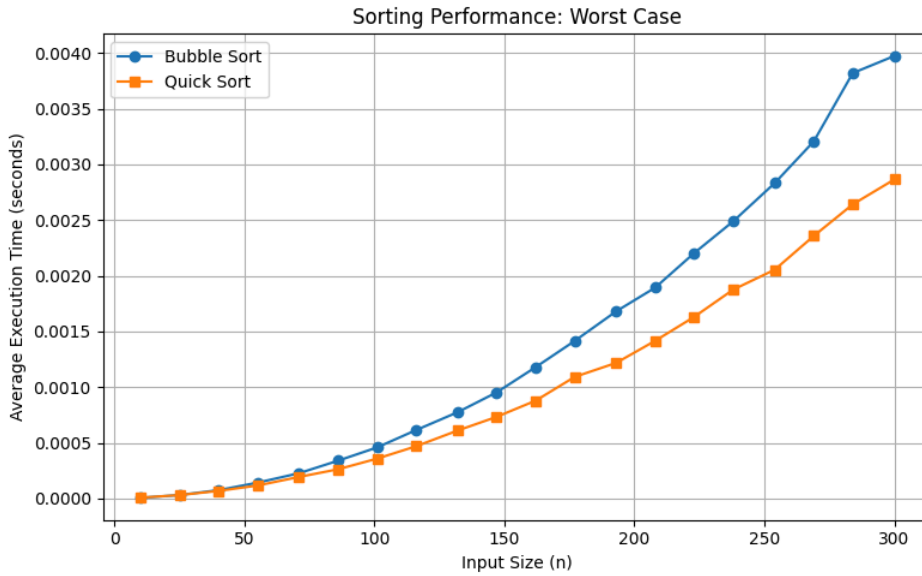


Ex2 - Questions

3. Generate performance plots for all the six (three?) cases. Performance of both algorithms must be visualized in each plot. In the plots, highlight for which inputs one algorithm performs better than the other.





4. Choose a threshold (on the input size) that determines whether the input is "small" or not. Justify your choice based on the plots.

We observe that below $n=30$, both algorithms run quickly enough that Bubble Sort's extra cost is negligible. As n grows beyond 30, Quick Sort begins to outperform Bubble Sort consistently. By the time n reaches 50, Bubble Sort's runtime grows rapidly compared to Quick Sort. Therefore, we choose $n=50$ as our threshold for "small" input sizes.